

Plant product residue monitoring is conducted as part of the National Residue Survey (NRS) by the Department of Agriculture, Fisheries and Forestry, through a grains residue testing program and three separate horticulture product residue testing programs.

Industry participation in these programs is voluntary and based on export and domestic market access and quality assurance objectives.

Outputs of NRS plant product residue monitoring include:

- provision of independent, authoritative and technically-sound residue data, reports and advice on Australian grain and horticultural products
- provision of residue monitoring data to support the specific market access requirements of participating industries
- reports of contraventions to the relevant state/territory government regulatory authorities within agreed timeframes
- The NRS maintains information on maximum residue limits (MRLs) that apply for Australia and major export markets for industries supported by the NRS. All analysis results are checked for compliance with Australian standards and relevant international MRLs
- coordination of the [Australian Grains Industry Post Harvest Chemical Usage Recommendations and Outturn Tolerances](#) document which details the industry-agreed residue tolerances for domestic and overseas markets.

All plant product residue monitoring programs are designed, operated and reviewed within agreed budgets by the NRS in consultation with peak industry bodies.

Grains program

The grains program covers 24 grain commodities:

- cereal grains (wheat, barley, oat, maize, sorghum, millet, rye, triticale)
- pulses (adzuki bean, chickpea, cow pea, pigeon pea, field pea, faba/broad bean, lentil, vetch, navy bean, mung bean, lupin and soybean)
- oilseeds (canola, sunflower, safflower, linseed).

And the milled fractions of:

- wheat (whole grain, flour, bran)
- wheat durum (whole grain, semolina, bran durum)
- soybean (whole grain, flour, kibble)
- rye (whole grain, flour, bran)
- maize (whole grain, flour and polenta).

The program comprises a range of sub-programs, which aim to cover all known export and domestic grain streams. These are:

- export: bulk export, and export containers or bags
- domestic: milled products, maltsters, oilseed crushers, stockfeed manufacturers, feedlots, and food processors.

The program is funded by a statutory 'farm-gate-value' levy. The grain levy also helps fund the [Grains Research and Development Corporation \(GRDC\)](#) and [Plant Health Australia](#).

To see the breakdown of the levy per grain product please refer to the [current levy charge rates](#).

Sample collection and analysis

Samples are collected at grain handling establishments in accordance with NRS protocols and procedures using NRS sampling equipment.

Export grain samples are collected at export terminals and container packing facilities as the grain is loaded onto ships or into containers.

Samples of milling grain and their milled fractions are collected from domestic mills on randomly selected dates.

Milled fraction samples, such as flour, semolina, polenta, bran and kibble, are collected as the whole grain and milled product. This way, the results provide information on the relative concentration of pesticide residues in each fraction.

Grain is also sampled on delivery to domestic grain processors such as stock feed manufacturers, maltsters, feedlots, and oilseed and food processors.

Grain samples are freighted overnight directly to the contract analytical laboratory.

Chemical screens

All grain samples are subjected to a multi-residue chemical screen covering a range of pesticides and environmental contaminants. According to agreed industry sampling rates, a percentage of samples are also randomly subjected to an additional herbicide screen.

To see the full list of chemicals that the NRS test for please go to the [publications page](#) and select the plant datasets section.

List of chemical screens used on grain samples

Screen	Chemicals	Examples
Multi-residue	Insecticides	Over 90 analytes including acephate, abamectin, bifenthrin, diazinon, malathion, pyrethrin and spinosad

	Fungicides	Over 55 analytes including azoxystrobin, boscalid, captan, iprodione, fludioxonil and propiconazole
	Herbicides	Over 90 analytes including atrazine, bromacil, clopyralid, isoxaben, norflurazon and simazine
	Organochlorines	aldrin and dieldrin, chlordane, DDT, endosulfan, endrin, HCB, heptachlor, lindane (gamma HCH) and mirex
Special herbicide	Herbicides	amitrole, chlormequat, dichlorprop-P, diclofop-methyl, diquat, fenoxaprop-ethyl, flamprop-M-methyl, fluazifop-p-butyl, glufosinate, glyphosate, haloxyfop, paraquat, quizalofop ethyl and quizalofop-p-tefuryl
Imidazolinone herbicides	Herbicides	imazamox, imazapic, imazapyr, imazaquin, imazethapyr
Metals	Elements	arsenic, cadmium, copper, lead, mercury and nickel

Grain program results summary 2015–16 to 2024–25

Year	Export Bulk Samples	Compliance (%)	Export Container Samples	Compliance (%)	Domestic Samples	Compliance (%)
2015-16	2,900	99.8	2,284	98.9	1,006	96.1
2016-17	4,166	99.8	3,755	99.0	892	97.3
2017-18	2,948	99.8	2,029	99.0	880	97.4
2018-19	2,426	99.6	979	98.3	854	98.2
2019-20	2,211	99.7	1,130	99.3	890	97.7
2020-21	3,256	99.9	1,313	98.7	876	98.6
2021-22	4,156	99.9	1,580	98.9	832	99.4
2022-23	5,032	99.5	1,167	96.3	825	98.6
2023-24	4,224	99.7	1,781	98.7	830	99.0
2024-25	3,995	99.8	1,443	99.0	851	99.4

Horticulture programs

The horticulture programs cover:

- apple and pear (pome fruit)
- almond
- macadamia.

Sample collection and analysis

Samples can be collected directly from growers, packing houses or city fruit and vegetable markets. Each sampling plan aims to establish a nation-wide spread of samples covering as many producers as is practicable each year.

All horticulture samples are freighted overnight to the contracted laboratory. If necessary, the laboratory then forwards samples to a second laboratory for additional analysis.

Chemical screens

All horticulture samples are subjected to a multi-residue screen which covers a range of fungicides, organophosphates, organochlorines, synthetic pyrethroids, herbicides and dithiocarbamates. The NRS, with the permission of the Apple and Pear industry, are offering optional heavy metals and microbiological testing on pome fruit samples.

The multi-residue screen has been developed in consultation with all participating horticulture industry peak bodies.

According to agreed industry sampling, if a site nominates for the tests, almonds can be subjected to additional special herbicide, phosphine and/or heavy metals screening.

All Macadamias will receive a multi-residue and heavy metals test and some samples will be randomly selected for a special herbicide and or a fosetyl test.

Apple program results summary 2015–16 to 2024–25

Year	Samples Compliance (%)	
2015-16	282	97.9
2016-17	248	98.0
2017-18	246	95.5
2018-19	245	98.4
2019-20	290	99.7
2020-21	271	96.3

2021-22	208	99.0
2022-23	167	97.0
2023-24	166	98.8
2024-25	170	96.5

Pear program results summary 2015–16 to 2024–25

Year	Samples Compliance (%)	
2015-16	94	95.7
2016-17	99	98.0
2017-18	85	90.6
2018-19	77	98.7
2019-20	75	100.0
2020-21	85	92.9
2021-22	54	98.1
2022-23	32	100.0
2023-24	47	100.0
2024-25	52	100.0

Macadamia program results summary 2015–16 to 2024–25

Year	Samples Compliance (%)	
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2015-16	135	100.0
2016-17	129	100.0
2017-18	123	100.0
2018-19	128	100.0
2019-20	110	100.0
2020-21	128	100.0
2021-22	126	100.0
2022-23	112	100.0
2023-24	137	100.0
2024-25	159	98.7

Almond program results summary 2015–16 to 2024–25

Year	Samples Compliance (%)	
2015-16	35	91.4
2016-17	96	87.5
2017-18	98	86.2
2018-19	70	98.7
2019-20	112	98.2
2020-21	100	98.0

2021-22	109	98.2
2022-23	126	94.1
2023-24	147	99.3
2024-25	138	100.0

Source: <https://www.agriculture.gov.au/agriculture-land/farm-food-drought/food/nrs/plant-product-testing>